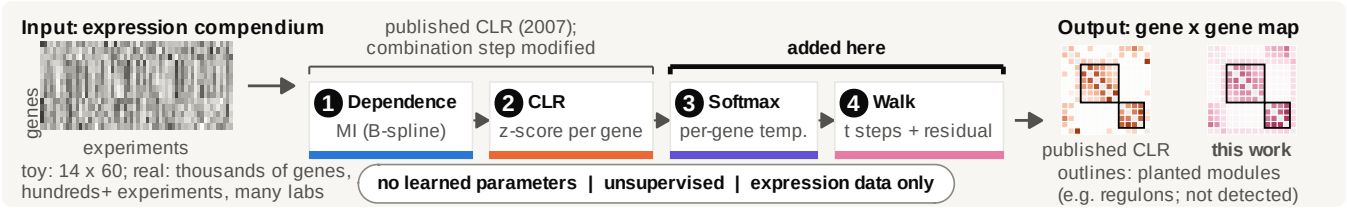
CLR as attention: compendium in, gene-association map out

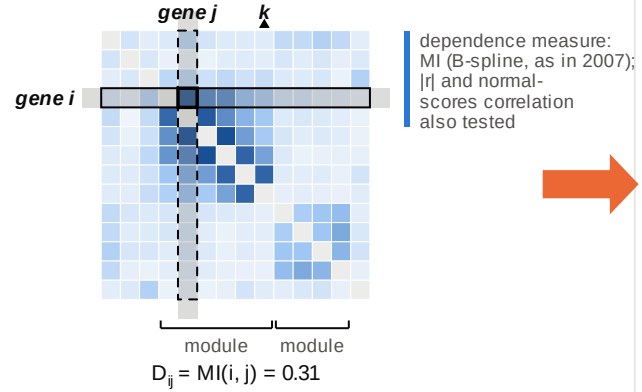
WORKED EXAMPLE (TOY): not results



1 Dependence

MI (B-spline, as in CLR 2007)

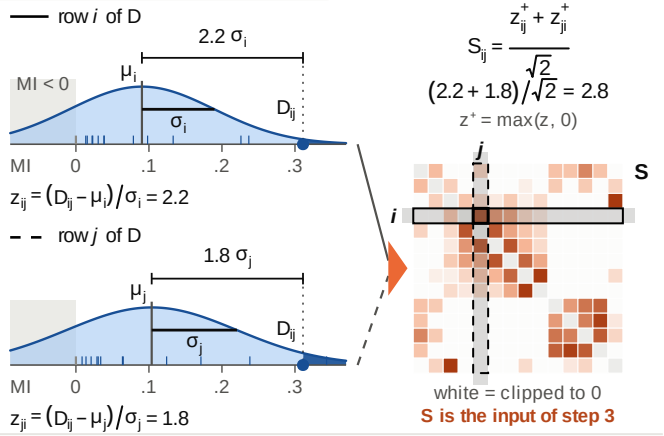
pairwise compatibility (parameter-free, symmetric)



2 Per-gene calibration (CLR)

z-scores as in CLR 2007; Stouffer (not Euclidean) here

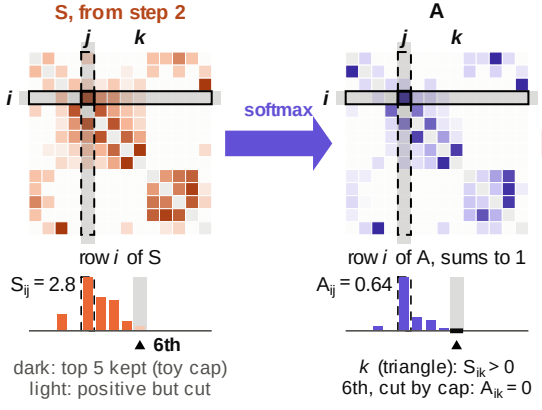
per-query normalization of the logits



3 Softmax attention

added here

softmax over the keys, temperature per query



$A_i = \text{softmax}(S_i/\tau_i)$; τ_i set so that $k_{eff} = 3$

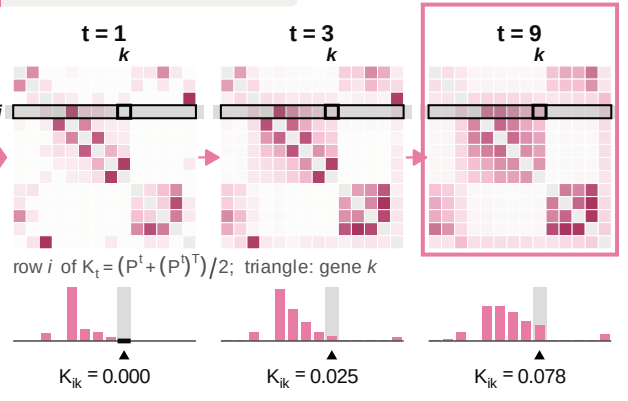
(toy k_{eff} 3, cap 5; real method: k_{eff} 10, top 50)

4 Walk: depth

added here

repeated attention with residual

$P = 0.5I + 0.5A$, repeated t times



toy: average precision (AP) of ranking same-module pairs; favourable by construction. Real data and losses: Figs. 2-6



framed: depth used by the method (chosen without labels on held-out experiments); the toy only displays it

Toy example; analogy with transformer attention, not equivalence; no real-data result shown.